obstacles present in front of it or near it. It helps in improving traffic conditions, provides relief to humans from daily travels, increases productivity of customers who would otherwise waste their time and energy in travelling back and forth, provides safety to those who are intoxicated and tired.

To understand exactly what an autonomous car does, The National Highway Traffic Safety Administration (NHTSA) explains various levels:

1stDegree Automation (no Automation): The driver is in complete control of all functions of the car like brake, throttle, steering, etc.

2ndDegree Automation (Function specific Automation): The driver has the control of the car but little control over few automatic functions like dynamic break during emergencies, cruise control, and lane keeping.

3rdDegree Automation (Combined Function Automation): The driver is freed of two primary control functions and one or two functions are automated.

4thDegree Automation (Limited Self Driving Automation): The driver has control over few small functions where the rest of the functions are automatic.

5thDegree Automation (Full Self Driving Automation): The car is entirely automatic and has control over all the functions and can completely drive it.

At this stage, Level 3 and Level 4 automobiles are being designed. However, it comes with its own challenges which need to be overcome in order to fully depend on this self driving car.

ANALYSIS OF RELATED WORKS

This study [1] used a reflexive neural network to run the autonomous car which converts the images into commands to drive it. The simulation data used to train the neural network was taken from Grand Theft Auto. The data set containing the necessary instructions was verified on connecting the neural network to the car system containing a camera input and the specific steering value, where the resulting average error rate was only 1.9%. This neural network was connected to a RC car system which provided a camera input and receives steering angle and throttling value.The designed car could successfully navigate 98 of the 100 laps in response to various obstacles placed in front of it. It consisted of aCNN network for image processing, a multi layered network for steering, a deep layered network for steering and additional networks for braking. It used python and Keras’ API to implement the neural network.The camera was used to provide images to the system and the existing steering angle as the input which was used to find the throttle. It consists of four modules: In Steering processing network, CNN provides a specific steering angle based on the input provided by the camera. It consists of series of layers that help in pattern detection.With merging network, the merged network provides the combined output of the previously described networks. It has a deep layered network of six layers.With Anti-collision network, this network is independent of the other networks and its main purpose is to make sure that the brakes are applied when an object comes in front of it.

The technology used in this [2] provided a solution to overcome monitoring of parking spaces using vision based automated parking. The current parking system just provided the number of